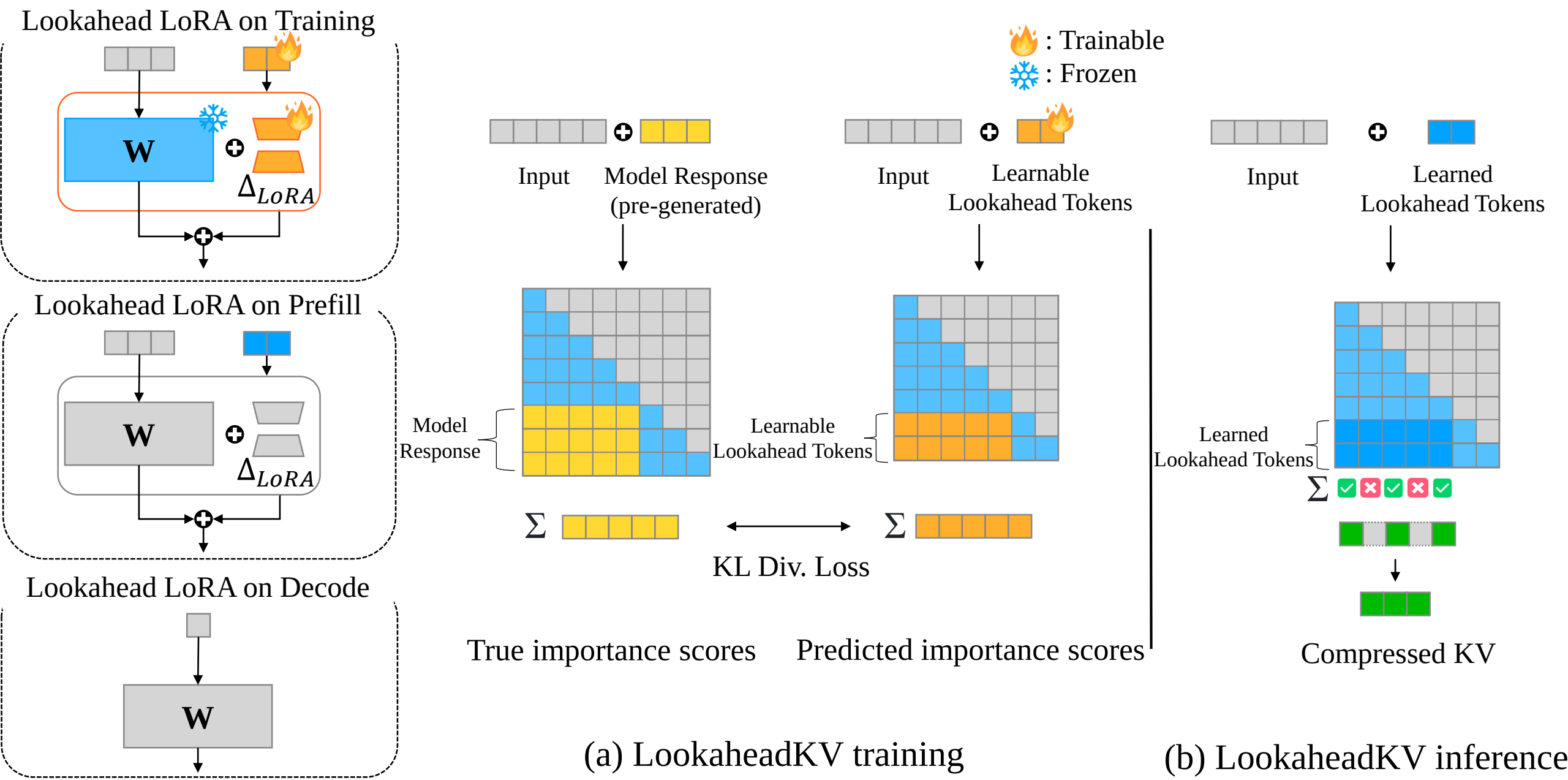
Lookahead LoRA on Training
W
Δ_{LoRA}
Lookahead LoRA on Prefill
W
Δ_{LoRA}
Lookahead LoRA on Decode
W
: Trainable
: Frozen
Input
Model Response
(pre-generated)
Input
Learnable
Lookahead Tokens
Input
Learned
Lookahead Tokens
Model
Response
Learnable
Lookahead Tokens
Learned
Lookahead Tokens
KL Div. Loss
True importance scores
Predicted importance scores
Compressed KV
(a) LookaheadKV training
(b) LookaheadKV inference